

Sets and elements

A **set** is an unordered collection of distinct objects. The objects in a set are its **elements**. For a set A and an object a ,

$$\begin{aligned} a \in A & \quad a \text{ is an element of } A, \\ a \notin A & \quad a \text{ is not an element of } A. \end{aligned}$$

Uppercase letters usually name sets; lowercase letters usually name elements. Two sets are equal if and only if they have the same elements.

Set-builder notation

$$\{x \mid x \text{ has property } P\}$$

denotes the set of all x with property P .

Common number sets

$\mathbb{N} = \{0, 1, 2, \dots\}$	natural numbers,
$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$	integers,
$\mathbb{Z}^+ = \{1, 2, \dots\}$	positive integers,
$\mathbb{Q} = \{p/q \mid p, q \in \mathbb{Z}, q \neq 0\}$	rational numbers,
$\mathbb{R}^+$	positive real numbers,
$\mathbb{C}$	complex numbers.

Empty and singleton sets

The **empty set** has no elements and is denoted by $\emptyset$ or $\{\}$. A **singleton set** has exactly one element.

Universal set

In a Venn diagram, the **universal set** U contains all objects under consideration.

Subsets and supersets

A is a **subset** of B , written $A \subseteq B$, if every element of A is also an element of B . Equivalently, B is a **superset** of A , written $B \supseteq A$.

$$A \subseteq B \iff \forall x (x \in A \rightarrow x \in B).$$

To show $A \not\subseteq B$, find an element of A that is not in B :

$$\exists x (x \in A \wedge x \notin B).$$

Theorem

For every set S , $\emptyset \subseteq S$ and $S \subseteq S$.

Proof

To show $\emptyset \subseteq S$, consider the statement $x \in \emptyset \rightarrow x \in S$. There is no $x \in \emptyset$, so the implication is vacuously true for every x . Also, $x \in S \rightarrow x \in S$ is true for every x , giving $S \subseteq S$.

Proper subsets

A is a **proper subset** of B , written $A \subsetneq B$, when $A \subseteq B$ and $A \neq B$.

Cardinality

If a set S contains exactly n distinct elements, where n is a nonnegative integer, then S is **finite** and its **cardinality** is $|S| = n$. For example,

$$|\mathbb{Z}| \text{ is infinite,} \quad |\{\emptyset, \mathbb{N}, \mathbb{Z}\}| = 3.$$

Power sets

The **power set** of S is the set of all subsets of S , denoted $\mathcal{P}(S)$. If $|S| = n$, then

$$|\mathcal{P}(S)| = 2^n.$$

Cartesian product

For sets A and B , their **Cartesian product** is the set of all ordered pairs whose first component is in A and second component is in B :

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

Example

The product of any set A with the empty set is empty:

$$\begin{aligned} A \times \emptyset &= \{(a, b) \mid a \in A \wedge b \in \emptyset\} \\ &= \emptyset, \end{aligned}$$

because no choice of $b \in \emptyset$ is possible.